

● HARD

● INFORMATION AND IDEAS

Inferences

10 questions · ~15 min

06

Q1

INFORMATION AND IDEAS

With the ongoing expansion of e-commerce, consumers are expecting faster and faster delivery of goods, but delivery companies continue to struggle with last-mile logistics (the final step in delivery to consumers) due to challenges such as complex and inefficient delivery routes. Innovations to mitigate these challenges have been emerging—the use of aerial drones, for example—but these innovations tend to engender their own complications (e.g., regulations on the use of drones in residential airspace), leading researchers to conclude that _____blank

Which choice most logically completes the text?

- A. consumers' expectations for reduced delivery times may be outstripping what is viable for delivery companies to provide.
- B. a better understanding of consumers' expectations for delivery is needed so that companies can better plan for fluctuations in delivery volume.
- C. rapid delivery is a leading factor in consumer satisfaction, and therefore delivery companies would benefit from investing resources in reducing delivery times.
- D. there may not be sufficient incentive for delivery companies to attempt to solve the problems associated with last-mile logistics.

Overgrazing by purple sea urchins has caused many kelp forests along North America's west coast to be replaced by urchin barrens—areas stripped of vegetation and covered in purple sea urchins. Urchins in barrens persist in a state of starvation that lessens their nutritional value—and thus their appeal—to many predators. Sarah Gravem and colleagues placed sunflower sea stars, a once-abundant predator species suffering massive population declines in recent years, in aquariums that each contained a nutritionally poor and a nutritionally rich purple sea urchin. The researchers found that the sea stars selected the nutritionally rich urchin in 42.7% of trials and the nutritionally poor urchin in 37.5% of trials, suggesting that _____ blank

Which choice most logically completes the text?

- A. sunflower sea stars are willing to hunt sea urchins, but if given a choice, they will prey on other more nutritious marine animals instead.
- B. sunflower sea stars are reluctant to feed on both nutritionally poor and nutritionally rich sea urchins and are therefore unlikely to thrive in kelp forests.
- C. sunflower sea stars are less likely to consume sea urchins in barrens than other species of sea stars are, putting sunflower sea stars at a high risk of extinction.
- D. sunflower sea stars do not always avoid foraging on nutritionally poor sea urchins, making sunflower sea star population recovery a potentially important tool for controlling urchin barrens.

The Uto-Aztecan language family is divided into a northern branch, which includes the Shoshone language of present-day Idaho and Utah, and a southern one, whose best-known representative is Nahuatl, the language of the Aztec Empire in Mexico. Lexical similarities across the family, including of botanical terms, confirm descent from a single language spoken millennia ago, and the family's geographical distribution suggests an origin in what is now the US Southwest. However, vocabulary pertaining to maize isn't shared between northern and southern branches, despite the crop's universal cultivation among Uto-Aztecan tribes. Given archaeological evidence that maize originated in Mexico and diffused northward into what became the US Southwest, some linguists reason that _____ blank

Which choice most logically completes the text?

- A. northern Uto-Aztecan tribes likely obtained the crop directly from a southern Uto-Aztecan tribe rather than from a non-Uto-Aztecan tribe.
- B. variation in maize-related vocabulary within each branch of the Uto-Aztecan family likely reflects regionally specific methods for cultivating the crop.
- C. southern Uto-Aztecan tribes likely acquired maize at roughly the same time as northern Uto-Aztecan tribes did, though from different sources.
- D. the family's division into northern and southern branches likely preceded the acquisition of the crop by the Uto-Aztecan tribes.

Marta Coll and colleagues' 2010 Mediterranean Sea biodiversity census reported approximately 17,000 species, nearly double the number reported in Carlo Bianchi and Carla Morri's 2000 census—a difference only partly attributable to the description of new invertebrate species in the interim. Another factor is that the morphological variability of microorganisms is poorly understood compared to that of vertebrates, invertebrates, plants, and algae, creating uncertainty about how to evaluate microorganisms as species. Researchers' decisions on such matters therefore can be highly consequential. Indeed, the two censuses reported similar counts of vertebrate, plant, and algal species, suggesting that _____ blank

Which choice most logically completes the text?

- A. Coll and colleagues reported a much higher number of species than Bianchi and Morri did largely due to the inclusion of invertebrate species that had not been described at the time of Bianchi and Morri's census.
- B. some differences observed in microorganisms may have been treated as variations within species by Bianchi and Morri but treated as indicative of distinct species by Coll and colleagues.
- C. Bianchi and Morri may have been less sensitive to the degree of morphological variation displayed within a typical species of microorganism than Coll and colleagues were.
- D. the absence of clarity regarding how to differentiate among species of microorganisms may have resulted in Coll and colleagues underestimating the number of microorganism species.

Among social animals that care for their young, such as chickens, macaque monkeys, and humans, newborns appear to show an innate attraction to faces and face-like stimuli. Elisabetta Versace and her colleagues used an image of three black dots arranged in the shape of eyes and a nose or mouth to test whether this trait also occurs in *Testudo* tortoises, which live alone and do not engage in parental care. They found that tortoise hatchlings showed a significant preference for the image, suggesting that _____ blank

Which choice most logically completes the text?

- A. face-like stimuli are likely perceived as harmless by newborns of social species that practice parental care but as threatening by newborns of solitary species without parental care.
- B. researchers should not assume that an innate attraction to face-like stimuli is necessarily an adaptation related to social interaction or parental care.
- C. researchers can assume that the attraction to face-like stimuli that is seen in social species that practice parental care is learned rather than innate.
- D. newly hatched *Testudo* tortoises show a stronger preference for face-like stimuli than adult *Testudo* tortoises do.

Volunteering, or giving time for a community service for free, is a valuable form of civic engagement because helping in a community is also good for society as a whole. In a survey of youths in the United States, most young people said that they believe volunteering is a way to help people on an individual level. Meanwhile, only 6% of the youths said that they think volunteering is a way to help fix problems in society overall. These replies suggest that _____ blank

Which choice most logically completes the text?

- A. many young people think they can volunteer only within their own communities.
- B. volunteering may be even more helpful than many young people think it is.
- C. volunteering can help society overall more than it can help individual people.
- D. many young people may not know how to find ways to volunteer their time.

One challenge when researching whether holding elected office changes a person's behavior is the problem of ensuring that the experiment has an appropriate control group. To reveal the effect of holding office, researchers must compare people who hold elected office with people who do not hold office but who are otherwise similar to the office-holders. Since researchers are unable to control which politicians win elections, they therefore _____ blank

Which choice most logically completes the text?

- A. struggle to find valid data about the behavior of politicians who do not currently hold office.
- B. can only conduct valid studies with people who have previously held office rather than people who presently hold office.
- C. should select a control group of people who differ from office-holders in several significant ways.
- D. will find it difficult to identify a group of people who can function as an appropriate control group for their studies.

Silicon-based photovoltaic cells account for 95% of the cells used in solar panels worldwide despite converting an average of only 18–22% of the sunlight that reaches them. In a study addressing this relative inefficiency, a team led by Laura Miranda-Pérez demonstrated that the addition of a thin layer of the mineral perovskite—which captures the blue range of light in the solar spectrum, whereas silicon captures the red range—allows the cells to convert 29.5% or more of the Sun’s energy into usable electricity. Cells made with only perovskite, however, are no more efficient than silicon-based ones. It’s reasonable to conclude, then, that _____ blank

Which choice most logically completes the text?

- A. photovoltaic cells with both silicon and perovskite are more efficient because they make use of more of the solar spectrum.
- B. photovoltaic cells with only perovskite and no silicon would likely convert more than 29.5% of the Sun’s energy.
- C. solar power will remain elusive until photovoltaic cells are replaced with a more practical technology.
- D. researchers need to evaluate whether other minerals like perovskite are as effective as perovskite seems to be.

Researchers Suchithra Rajendran and Maximilian Popfinger modeled varying levels of passenger redistribution from short-haul flights (flights of 50 to 210 minutes, from takeoff to landing) to high-speed rail trips. Planes travel faster than trains, but air travel typically requires 3 hours of lead time for security, baggage handling, and boarding that rail travel doesn't, so short-haul routes take similar amounts of time by air and by rail. However, the model suggests that as rail passenger volumes approach current capacity limits, long lead times emerge. Therefore, for rail to remain a viable alternative to short-haul flights, _____ blank

Which choice most logically completes the text?

- A. rail systems should offer fewer long-haul routes and airlines should offer more long-haul routes.
- B. rail systems may need to schedule additional trains for these routes.
- C. security, baggage handling, and boarding procedures used by airlines may need to be implemented for rail systems.
- D. passengers who travel by rail for these routes will need to accept that lead times will be similar to those for air travel.

German theater practitioner Bertolt Brecht (1898–1956) believed that theater should elicit an intellectual rather than an emotional response from audiences, provoking them to consider social and political realities that extend beyond the characters and events depicted onstage. Brecht's influence can be seen in English playwright Caryl Churchill's 1979 play *Cloud 9*: although the play sometimes invites empathetic reactions, it primarily works to engage audiences in an interrogation of patriarchy and colonialism, which it does by placing audiences at a distance, thereby encouraging them to _____ blank

Which choice most logically completes the text?

- A. focus on the characters' beliefs about social and political issues as revealed by the characters' actions.
- B. reflect on social and political phenomena not directly related to patriarchy and colonialism.
- C. recognize pertinent social and political parallels between Germany during Brecht's time and England at the time when Churchill was writing *Cloud 9*.
- D. be dispassionate as they think critically about the social and political questions raised by the play.